ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. EE ENGINEERING — SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Derive EMF equation of DC Generator. Explain Armature Reaction and remedies. [14 Marks]

[OR]

Q.1 (Alt) Compare Lap and Wave windings. Explain commutation and interpoles. [14 Marks]

--- UNIT II ---

Q.2 Derive Torque equation of DC motor. Explain speed control methods of DC motors. [14 Marks]

[OR]

Q.2 (Alt) Explain Swinburne's test and Hopkinson's regenerative test. Describe 3-point starter. [14 Marks]

--- UNIT III ---

Q.3 Draw exact equivalent circuit of single-phase transformer. Explain OC and SC tests. [14 Marks]

[OR]

Q.3 (Alt) Derive condition for maximum efficiency of transformer. Explain Voltage Regulation. [14 Marks]

--- UNIT IV ---

Q.4 Explain Autotransformer copper savings. Describe parallel operation of transformers. [14 Marks]

[OR]

Q.4 (Alt) Explain Scott Connection (T-T connection) for 3-phase to 2-phase conversion. [14 Marks]

--- UNIT V ---

Q.5 Explain Sumpner's Back-to-Back test for identical transformers efficiency. [14 Marks]

[OR]

Q.5 (Alt) Describe tap-changing transformers (on-load and off-load) and cooling methods. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****